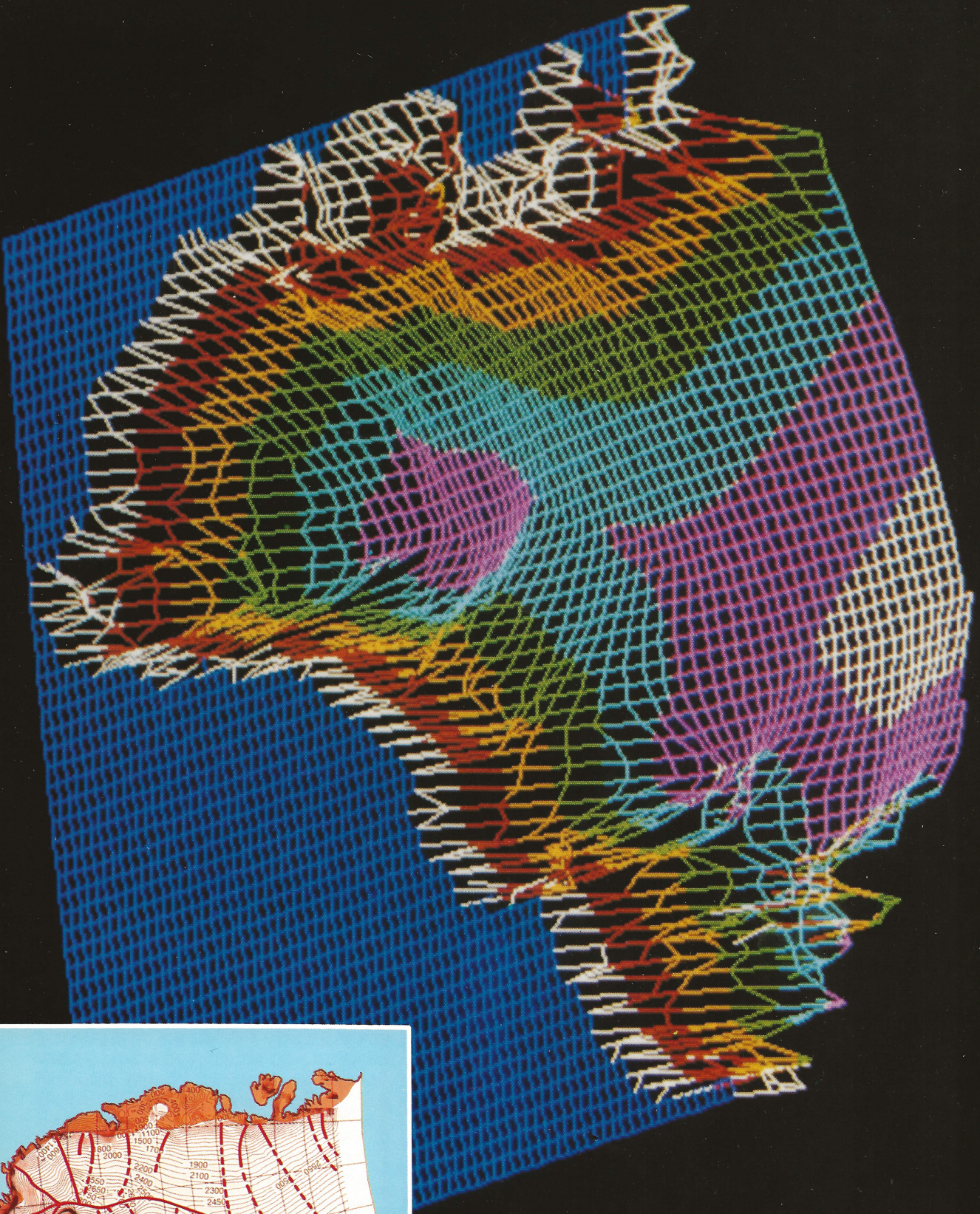
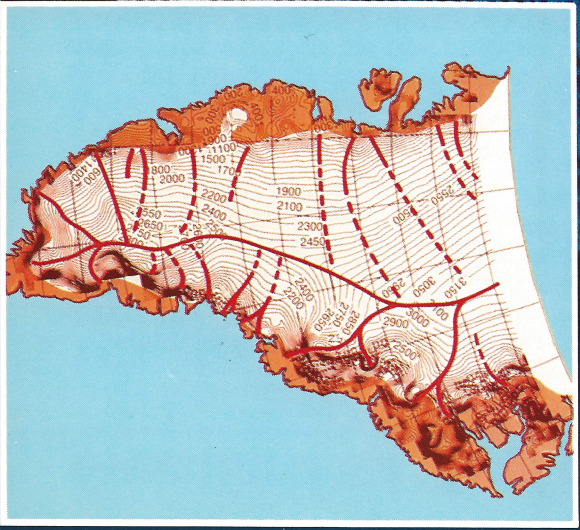


# Greenland Ice Sheet





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Greenland is covered by a remnant of the great ice sheets that have periodically covered much of the northern hemisphere during the past few million years. The last worldwide glacial advance (or "ice age") lasted almost 100,000 years; about 15,000 years ago it ended when warming temperatures triggered a general retreat. We are now in an "interglacial," a warm period between glaciations which generally lasts about 20,000 years. Although the approximate dates for various glacial cycles have been identified, neither the mechanisms that trigger the beginning and end of an ice age, nor the rate at which the Earth's climate flips between glacial and interglacial conditions are known. Consequently, we cannot say whether and when a new advance of the glaciers will occur.

The natural cycle of ice ages is further complicated by man's activities. Burning of fossil fuels and deforestation of the land are contributing to a steady increase in atmospheric carbon dioxide ( $\text{CO}_2$ ). (See: Gulf Stream Temperature.) Given that  $\text{CO}_2$  can trap heat within the atmosphere, a global warming has been predicted that could seriously alter the present configuration of the ice sheets. At high latitudes, ice and snow reflect much of the sun's energy away from the Earth. Consequently, a climate change causing a reduction in the ice and snow cover can lead in turn to further warming, as solar heating is able to penetrate previously covered land and ocean surfaces. Indeed, climate models indicate that such a warming will be significantly amplified in the polar regions. Eventually, sufficient warming could affect even the great ice sheets, causing them to shrink with an accompanying rise in sea level.

During the past 100 years, sea level has risen perhaps 10 to 15 cm (four to six inches). Part of this rise is probably due to the melting and retreat of glaciers exclusive of those in Greenland and Antarctica. But, despite 25 years of intensive field work, it is still not known whether the Greenland and Antarctic ice sheets are growing or shrinking. In Greenland, for example, there have been repeat measurements of only one leveling traverse across the ice sheet. Results indicate a thinning of the ice at low elevations, with a thickening further up slope, but these trends are unlikely to be representative of the entire ice sheet. Moreover, not a single traverse across Antarctica has been leveled even once, and the surface topography in most areas is known only poorly. Conventional survey techniques, even airborne measurements, will never provide data at a sufficiently high spatial density over entire ice sheets. For such data, we must take advantage of satellite altimetry.

The radar altimeter aboard NASA's Seasat obtained almost a million useful measurements of surface elevation over Greenland and Antarctica during its three-month lifetime in 1978. Measurement consistency over smoother portions of the ice sheets was a few tenths of a meter, with increasing errors over sloping or undulating surfaces.

Data collected over Greenland were analyzed by Jay Zwally and Robert Bindshadler at the Goddard Space Flight Center and are displayed here both in a three-dimensional perspective and in a more traditional topographic map, revealing the overall shape of the southern half of Greenland. (Because the Seasat orbit limited data acquisition to the area between 72° N and 72° S latitude, there is no coverage of the northern half of Greenland.) Ice covers the entire region except for the coastal mountains. These are breached by outlet glaciers flowing into the sea, many of which can be identified by valleys in the ice sheet surface farther upstream. Surface elevations range from near sea level to more than 3250 m (10,700 ft) in the central region, where ice thickness exceeds 3500 m (11,500 ft). In the three-dimensional perspective, the vertical scale is exaggerated by a factor of 150 to give a better impression of ice sheet shape; each 500-m (1650-ft) band of elevation is marked with a different color.

Glaciers and ice sheets flow due to gravity, with the force being proportional to the surface slope, which can readily be derived from satellite altimeter data. Moreover, maps of the magnitude and direction of surface slope delineate drainage basins, as well as provide an indication of conditions beneath the ice sheets, such as bedrock topography. This information is essential to understanding the behavior of an entire ice sheet, and it enables identification of areas of particular interest that merit closer investigation by airborne or ground-based surveys. But the most exciting prospect offered by satellite altimetry is the unambiguous detection of regions within an ice sheet where the ice is either thickening or thinning. The US Navy GEOSAT, launched in 1985 into an orbit covering a latitude band similar to that of Seasat, is acquiring radar-altimeter data which should reveal any significant changes that have occurred since 1978. Then, in late 1989, areal coverage will be extended to 82° N latitude by the altimeter aboard the European Space Agency's ERS-1 satellite. Such repeat surveys at decadal intervals could provide an early warning of ice sheet changes that might herald a period of warming climate and rising sea levels.



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